

Customer Churn Prediction Using Machine Learning

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Abstract

- ▶ Predicts customer churn using Machine Learning.
- ▶ Uses customer information such as **gender, tenure, contract type, monthly charges, and total charges**.
- ▶ Performs **data preprocessing, label encoding, and feature scaling** to improve model performance.
- ▶ Trains and compares **multiple Machine Learning algorithms** to identify the best model.
- ▶ Selects the **highest-accuracy model** for churn prediction.
- ▶ Deploys the trained model using a **Streamlit web application**.
- ▶ Provides **real-time churn predictions** through a simple and user-friendly interface.
- ▶ Helps businesses improve **customer retention**, reduce churn, and support **data-driven decision-making**

Problem Statement

- ▶ High customer churn leads to revenue loss and reduced customer retention.
- ▶ Traditional prediction methods are time-consuming.
- ▶ Businesses need early identification of high-risk customers.
- ▶ Machine Learning enables accurate churn prediction.
- ▶ Early prediction improves customer retention.

Objectives

- ▶ Predict customer churn using customer demographic and service-related data.
- ▶ Develop a **Random Forest** model to accurately identify customers who are likely to discontinue the service.
- ▶ Improve prediction performance through data preprocessing techniques such as feature encoding and feature scaling.
- ▶ Build an interactive **Streamlit** web application for real-time customer churn prediction.
- ▶ Support business decision-making by enabling early identification of high-risk customers and improving customer retention strategies.

Dataset Description



Dataset Overview

The dataset used for Customer Churn Prediction contains **7043 rows** and **21 columns**.



Important Features

- CustomerID
- Gender
- SeniorCitizen
- Partner
- Dependents
- tenure (months)
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling
- PaymentMethod
- MonthlyCharges
- TotalCharges



Target Variable

Churn – Indicates whether the customer churned or not.
(Yes / No)

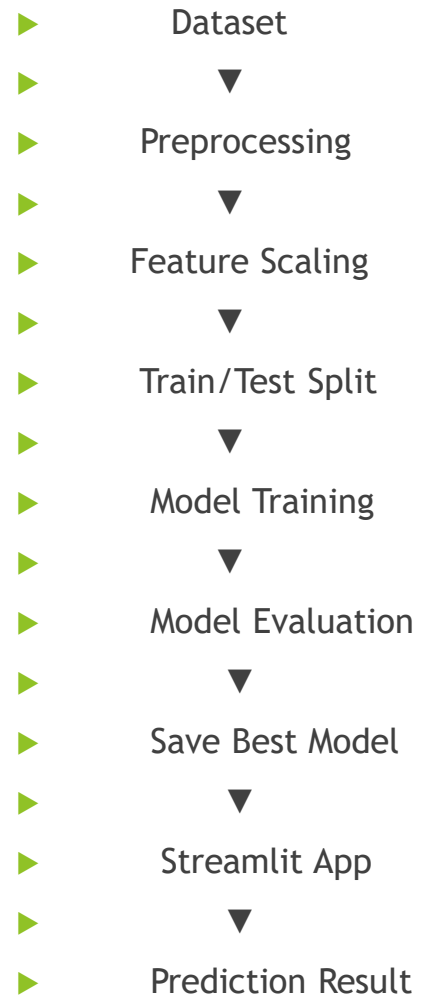


Dataset Summary

Total Records	Total Features	Numerical Features	Categorical Features
7043	21	7	14



Project Architecture



Software Requirements

Software		Specification
Operating System	→	Windows 10/11
Programming Language	→	Python 3.10 or above
IDE	→	Jupyter Notebook / Visual Studio Code
Libraries	→	Pandas, NumPy, Scikit-learn, Streamlit, Joblib, Matplotlib
Database/File Format	→	CSV (Customer Churn Dataset)
Browser	→	Google Chrome, Microsoft Edge

Actual Outcome

- ▶ Successfully developed a machine learning model to predict whether a customer is likely to churn based on customer details.
- ▶ Performed data preprocessing, including encoding categorical features and scaling numerical data, to improve model performance.
- ▶ Trained and evaluated multiple machine learning algorithms and selected the best-performing model for prediction.
- ▶ Built a user-friendly **Streamlit** web application that allows users to enter customer information and instantly view the churn prediction.
- ▶ The system helps businesses identify potential churn customers in advance, enabling better customer retention strategies and data-driven decision-making.

Future Scope

- ▶ Enhance prediction accuracy by training the model on larger and more diverse customer datasets.
- ▶ Implement advanced machine learning and deep learning algorithms to improve prediction performance.
- ▶ Include additional customer features such as contract type, internet service, payment method, and customer support history.
- ▶ Deploy the system on cloud platforms and integrate it with CRM systems for real-time customer churn monitoring.
- ▶ Develop a mobile-friendly application with interactive dashboards for business managers to access churn predictions anywhere.

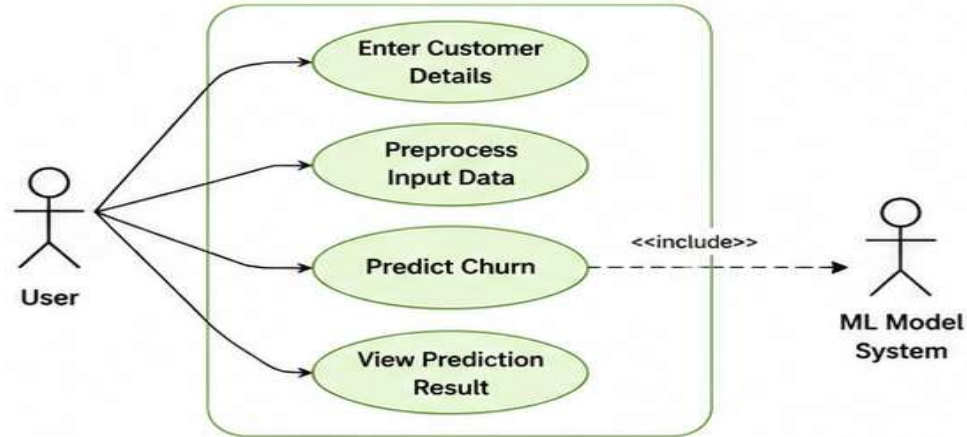




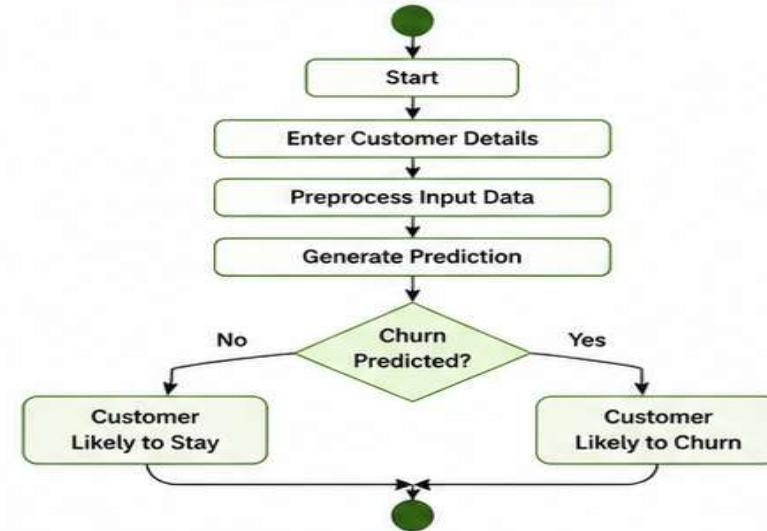
UML DIAGRAMS



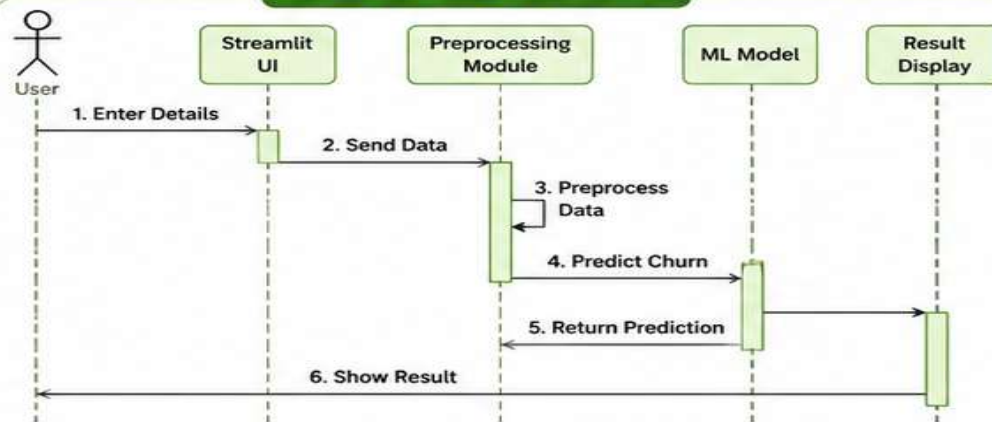
1. USE CASE DIAGRAM



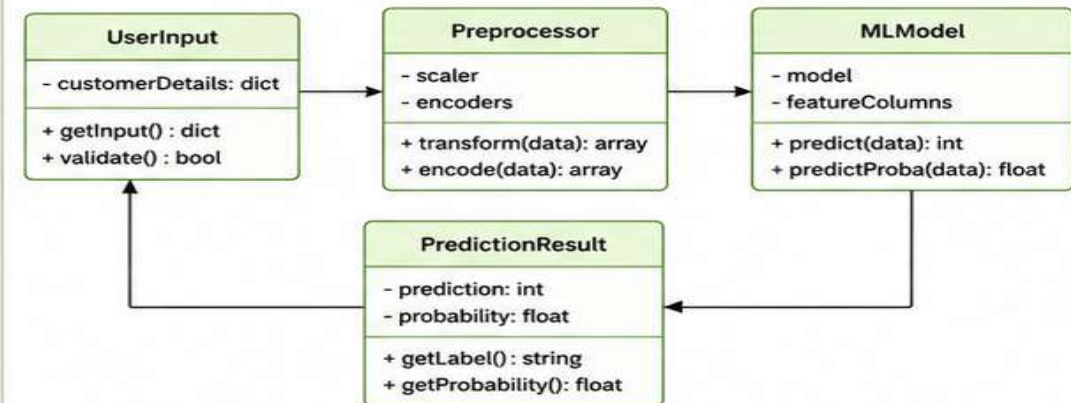
2. ACTIVITY DIAGRAM



3. SEQUENCE DIAGRAM



4. CLASS DIAGRAM



Visualizing the structure and flow of the Customer Churn Prediction System



Better understanding of system components and their interactions



Helps in analysis, design and implementation of the system

Result / Streamlit Web Application

Customer Churn Prediction

Enter a customer's details below and the model will predict whether they're likely to churn.

Customer details

gender Female	Partner No
Dependents No	PhoneService No
MultipleLines No	InternetService DSL
OnlineSecurity No	OnlineBackup No
DeviceProtection No	TechSupport No
StreamingTV No	StreamingMovies No
Contract Month-to-month	PaperlessBilling No
PaymentMethod Bank transfer (automatic)	SeniorCitizen No
tenure (months) 12	MonthlyCharges 89.00
TotalCharges 50000.00	

Predict Churn

This customer is likely to stay. (Churn probability: 35.0%)

- Developed a user friendly Streamlit app.
- Accepts customer input.
- Processes input using Standard Scaler.
- Predicts customer churn
- Displays result instantly.

- Customer Churn Prediction Web Application

Conclusion

- ▶ Successfully built a **Machine Learning-based Customer Churn Prediction System**.
- ▶ Preprocessed customer data and trained multiple classification models for accurate prediction.
- ▶ Evaluated different machine learning algorithms and selected the best-performing model.
- ▶ Developed a **Streamlit** web application that provides an easy-to-use interface for predicting customer churn.
- ▶ The proposed system helps businesses identify customers who are likely to churn, enabling better customer retention strategies, reducing customer loss, and supporting data-driven business decisions.

THANK YOU!